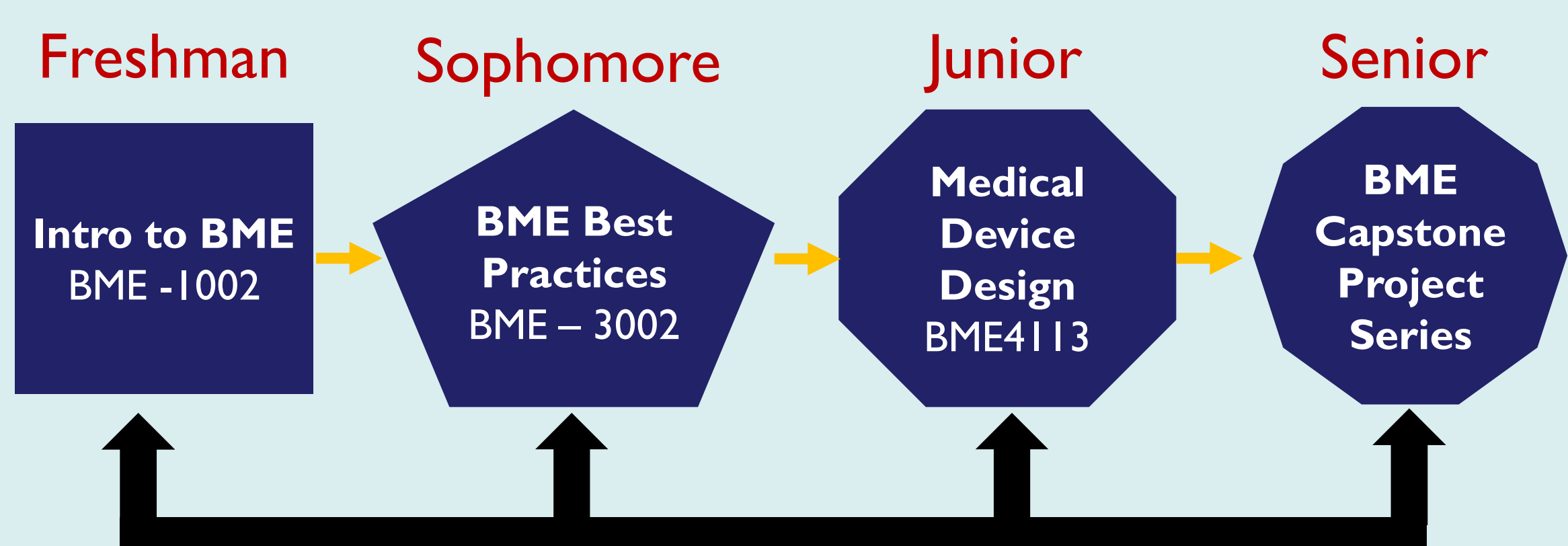


Biomedical Engineering

Introducing Entrepreneurship to Biomedical Students at various points in their education using currently relevant technologies



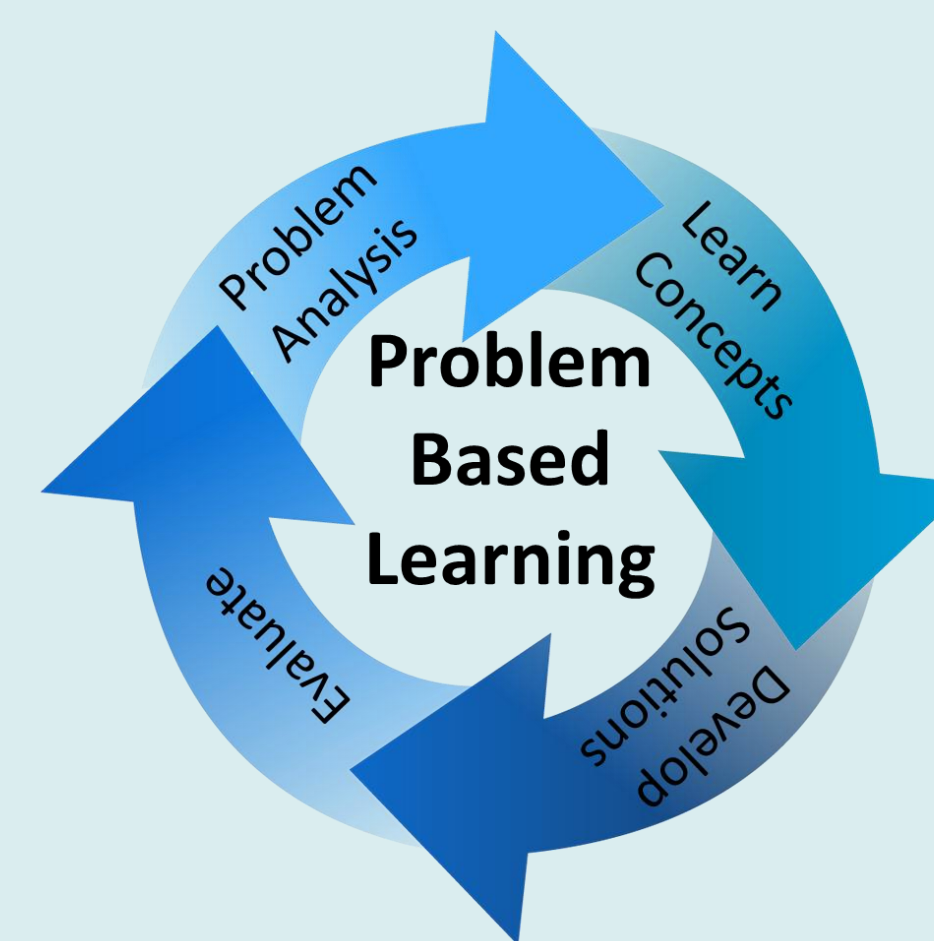
Entrepreneurial Experiences

	Opportunity Recognition	Market Investigation	Create a Preliminary Model	Communicate solutions in terms of economic benefits	Communicate solutions in terms of societal benefits	Examine technical feasibility, economic drivers, societal and individual needs	Intellectual Property Protection	Regulatory Issues	Collaborate in a team setting
Intro. to BME	X	X		X	X				X
BME Best Practices		X		X	X		X	X	X
Medical Device Design	X	X	X		X	X		X	
BME Capstone Series	X	X	X	X	X	X	X	X	X

Problem-Based Learning

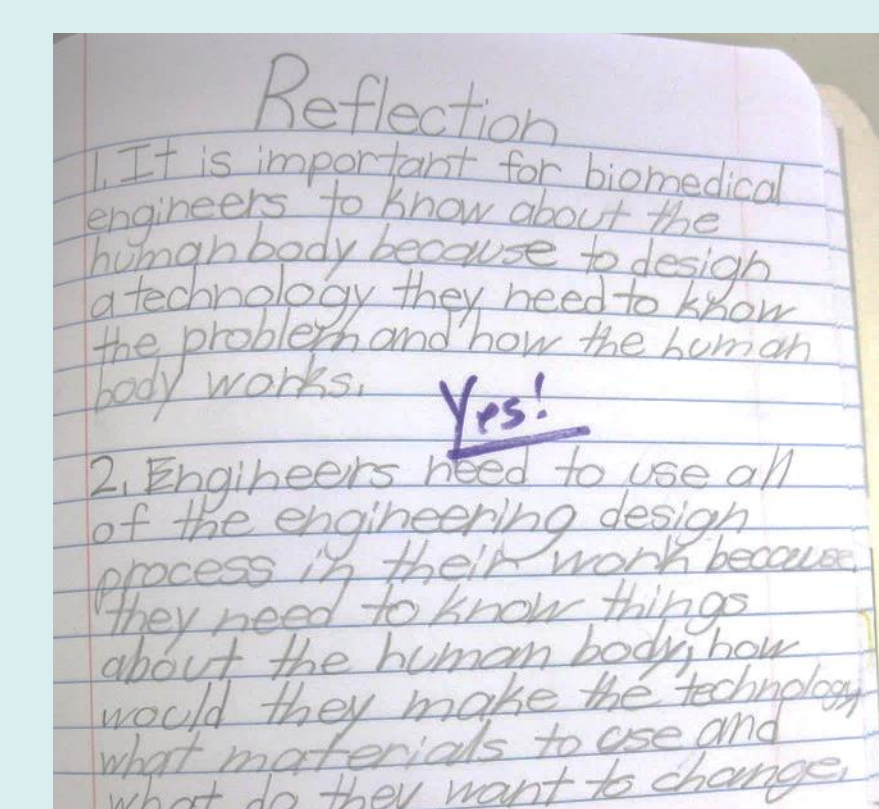
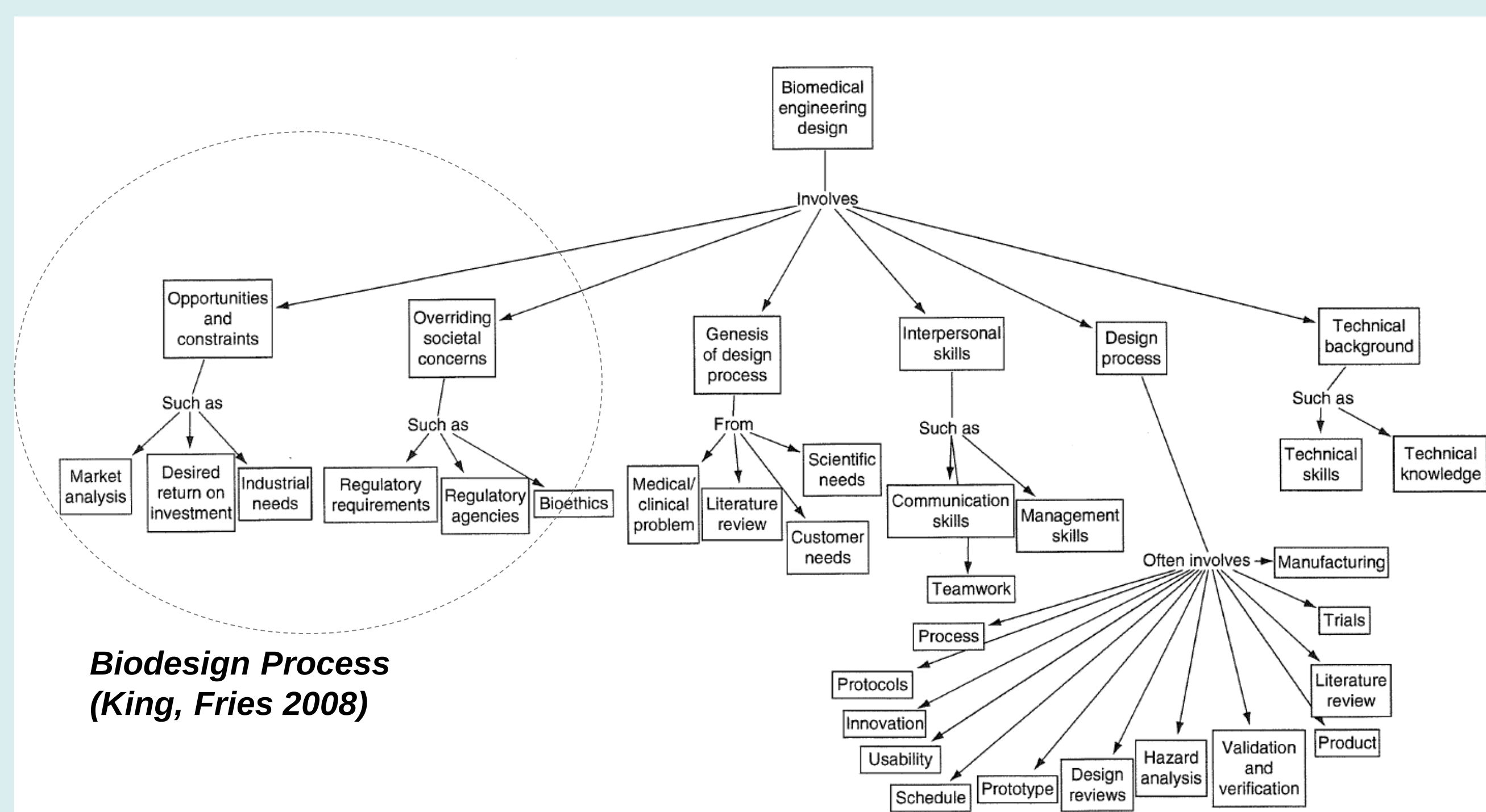
This is what is normally taught in classes

Problems Given
↓
Techniques Taught
↓
Solve



This is what is real world problems look like

Problems / Needs
↓
Many Constraints
↓
Many Tools
↓
Many Solutions



Curiosity
Connections
Creating Value

- What is the BME Advantage?**
- Understand the Design Process
 - Larger tool set and skills
 - Bridge the knowledge of biology and engineering

Entrepreneurial Minded Learning

Focus

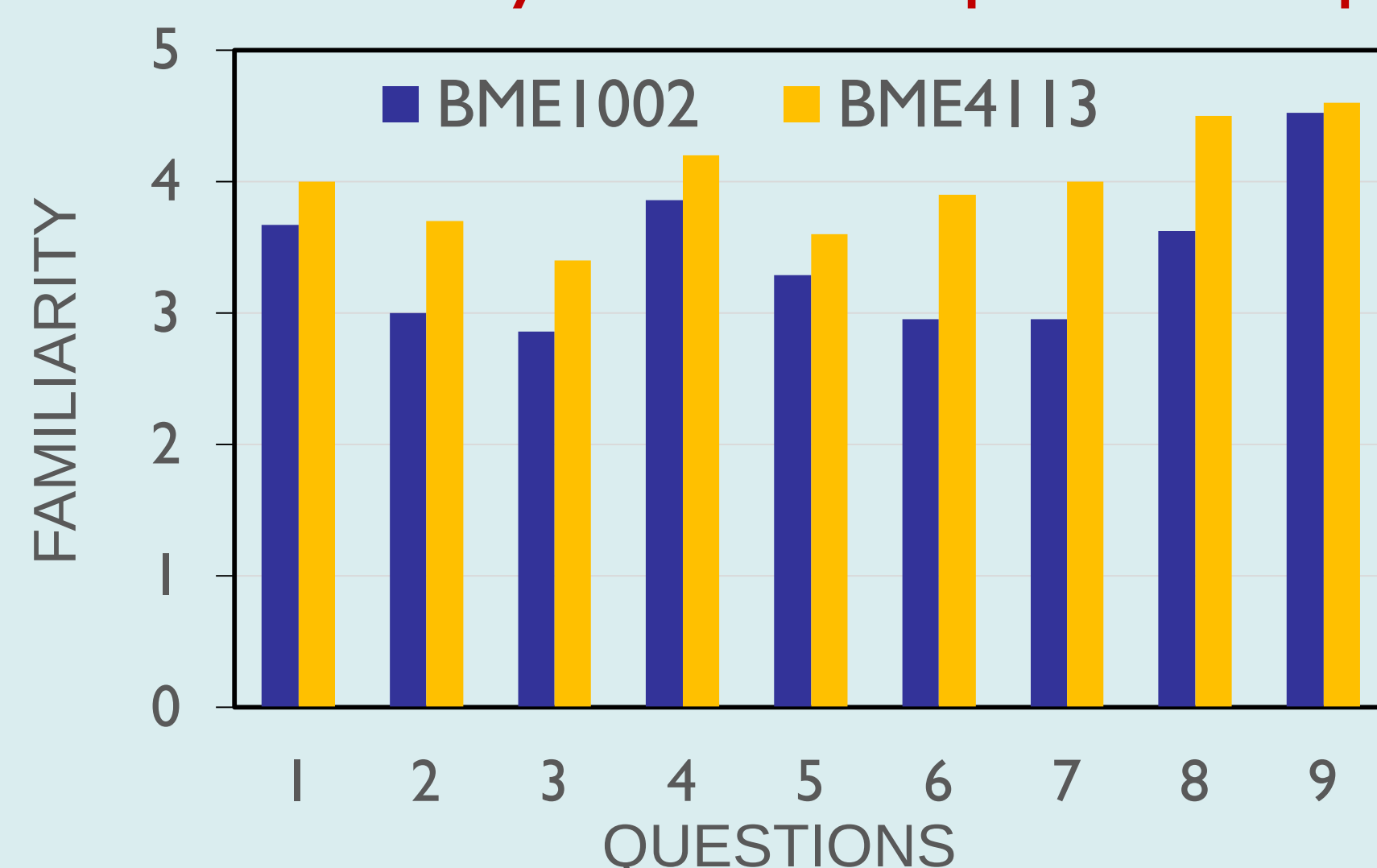
- Opportunity Recognition
- Customer Interaction



Implementation Strategy

- Faculty Buy-in
- Common Platform
- Develop Resources / Modules
- Implement in Classroom
- Assessment
- Propose Modifications
- Disseminate Results

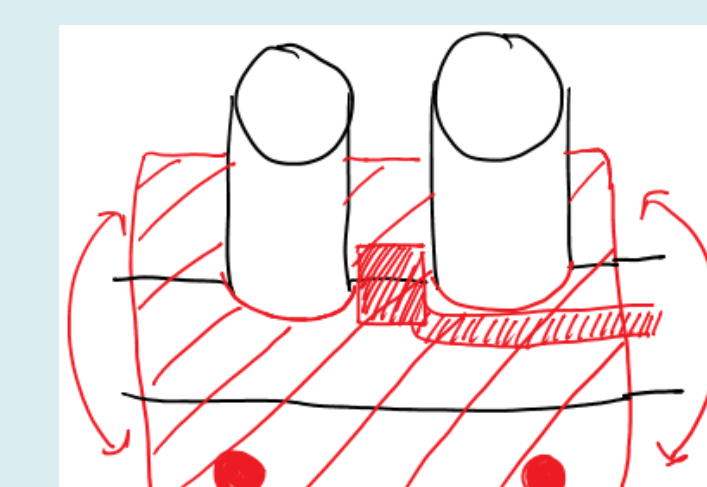
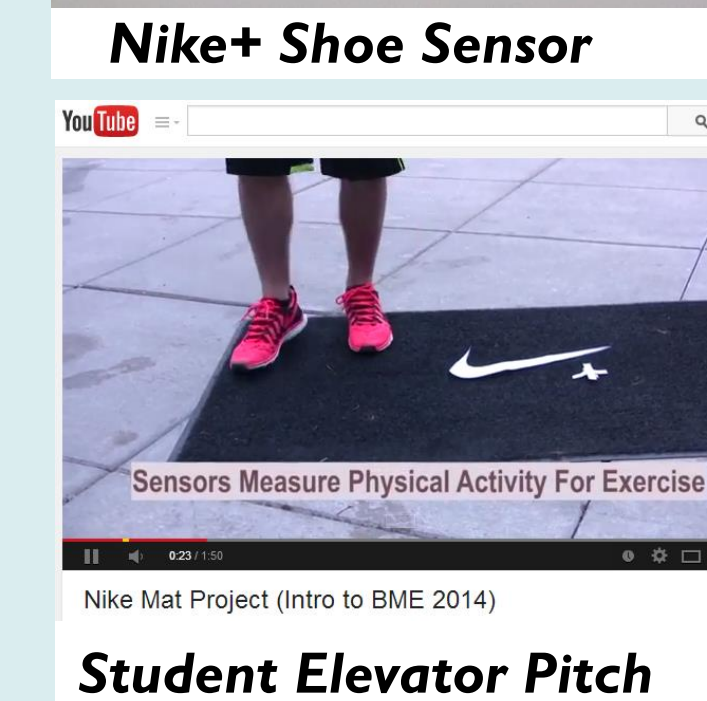
Familiarity with Entrepreneurship



This project was supported by Kern Family Foundation Topical Grant

Sample Activities and Student Output

Bisociation / Painstorming Activity	General
Repurpose 510k as a QS Device	Intro to BME Best Practices
Business Model / Pitch	Intro to BME
MedTech Familiarity	Medical Device Design
Needs Finding / Opportunity	Medical Device Design
Improvement to Existing Technology	Orthopedics



Moving Forward

- How effective is it?
- Developing Validated Instruments of Assessment
- Finding a skill balance → Technical vs Entrepreneurial